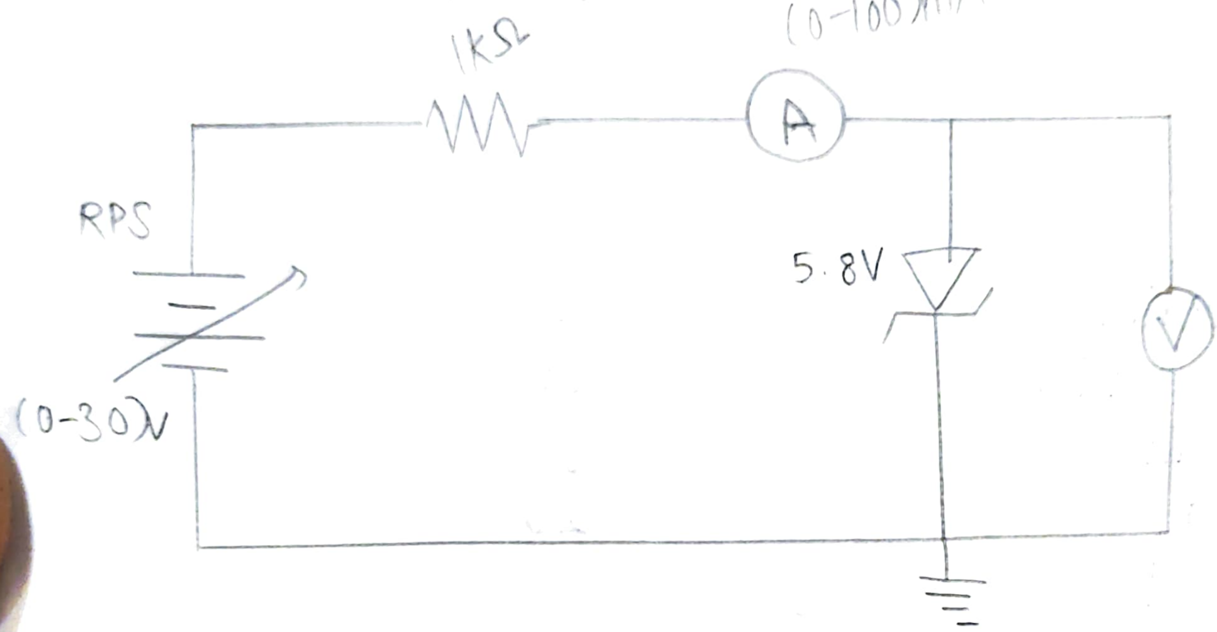
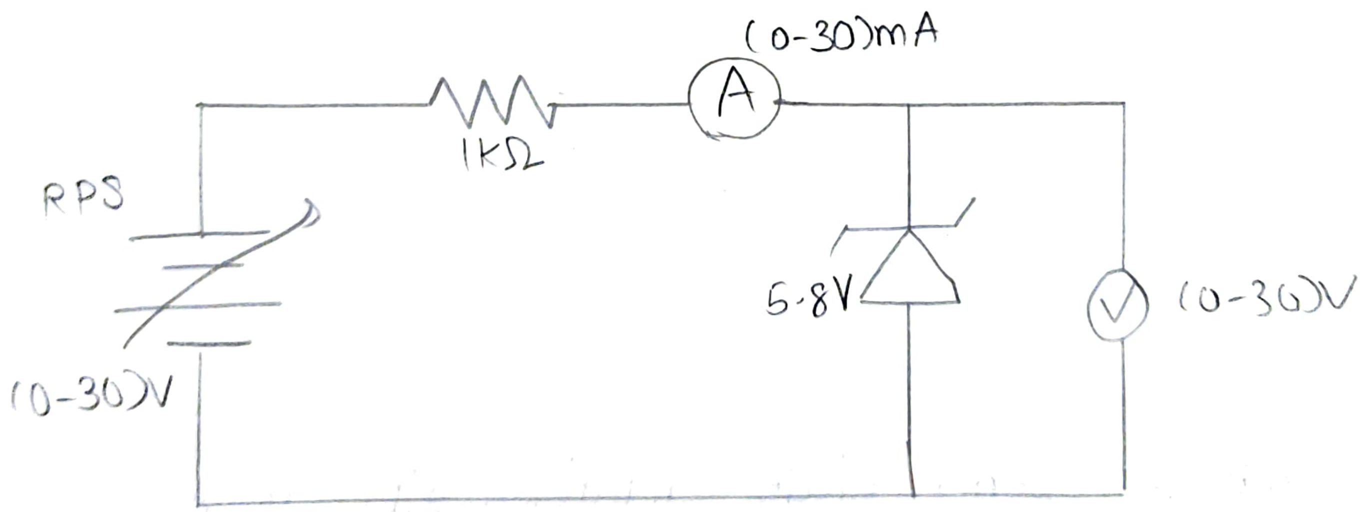
(1st exp)

Forward bias,



Reverse Bias,



TABULATION:-

Forward Bias		Reverse Bias	
Voltage (V)	Current (mA)	Voltage (V)	Current (mA)
0.1	0	0	0
0.2	0	1	0
0.3	0	2	0
0.4	0	3	0
0.5	0	4	0
0.6	0	5	0
0.7	0.86	5.11	1.37
0.71	1.09	5.12	1.87
0.72	1.60	5.13	2.29
0.73	2.53	5.14	4.07
0.74	3.30	5.15	6.57
0.75	4.61	5.16	8.19
0.76	6.82	5.16	10.05
0.77	9.42	5.16	13.03
0.78	12.82	5.16	15.26
0.79	15.52	5.16	17.44
0.80	20	5.16	18.24

Expt. No.

Page No.

(v) The breakdown voltage is noted before the breakdown and connection is changed such that lower range ammeter is included in the circuit.
(vi) Graph is drawn between V_r and I_r.
(vii) The reverse conduction region is extended to meet at point. The value at this point is V_{rb}.

FORMULA:-

Forward resistance,

$$R_F = \Delta V_F / \Delta I_F \; (\Omega).$$

Reverse resistance,

$$R_R = \Delta V_R / \Delta I_R \; (\Omega)$$

CALCULATION:-

Forward resistance,

$$R_F = \frac{\Delta V_F}{\Delta I_F} \ (\Omega)$$

$$= \frac{(0.74 - 0.73)}{(3.30 - 2.53)} \times 10^3$$

$$= \frac{0.01}{0.77} \times 10^3 \Rightarrow 0.012 \times 10^3$$

$$\Rightarrow R_F = 12\Omega.$$

Reverse resistance,

$$R_r = \frac{\Delta V_r}{\Delta I_r} \ (\Omega)$$

$$= \frac{(5.12 - 5.11)}{(1.87 - 1.37)} \times 10^3$$

$$= \frac{0.01 \times 10^3}{0.50} \Rightarrow \frac{10.}{0.5} \Rightarrow 20$$

$$\Rightarrow R_r = 20\Omega.$$